

Material Safety Data Sheet

Section I. Chemical Product and Company Identification

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PRODUCT NAME: **PermaPlate Leatherguard®**

IMPORTANT! Read this MSDS before use or disposal of this product. Pass along the information to employees and other persons who could be exposed to the product to be sure that they are aware of the information before use or other exposure. This MSDS has been prepared according to the OSHA Hazard Communication Standard [29 CFR 1910.1200]. The MSDS information is based on sources believed to be reliable. However, since data, safety standards, and government regulations are subject to change and the conditions of handling and use, or misuse are beyond our control, SISKIN ENTERPRISES, INC. makes no warranty, either expressed or implied, with respect to the completeness or continuing accuracy of the information contained herein and disclaims all liability for reliance thereon. Also, additional information may be necessary or helpful for specific conditions and circumstances of use. It is the user's responsibility to determine the suitability of this product and to evaluate risks prior to use, and then to exercise appropriate precautions for protection of employees and others.

Section II. Composition / Information on Ingredients

2.1 Chemical characterization (preparation):

Chemical characteristics
Polydimethylsiloxane (emulsion in water)

2.2 Information on ingredients:

Type	CAS No.	Substance	Content [wt. %]		Note
			Lower	Upper	
INHA	556-67-2	Octamethyl cyclotetrasiloxane	0.1	<1.0	R
INHA	67923-07-3	Amino functional polydimethyl siloxane	1.0	5.0	
INHA	9016-45-9	Nonylphenyl hydroxy polyglycol ether	1.0	5.0	
INHA	9036-19-5	Polyoxyethylene (t-octylphenyl) ether	1.0	5.0	

Type: HYD - INHA – ingredient

Note: R - reproductive toxin.

Substances listed in the Subsections "HAPS" and "California Proposition 65 Carcinogens / Reproductive Toxins" that are not listed in Section 2 are only present at quantities below 0.1% for California Proposition 65 listed toxins or below 1% for non-carcinogenic HAPS or they are inextricably bound in the product.



Section III. Hazards Identification

3.1 Hazards classifications

HMIS® rating (product as packaged):

Health: 2*

Fire: 0

Reactivity: 0

PPE: H

Note: Respiratory protection is only recommended in the event that ventilation or engineering controls are unable to maintain exposures below recommended levels; or in the event of a spill or other emergency response situation. (HMIS codes are based on contact with the product as packaged and any hydrolysis by-products, if present.) Hazardous Materials Identification System and HMIS are registered trademarks of the National Paint and Coatings Association

Canadian WHMIS Classification: D2A, D2B

3.2 Emergency Overview and Potential Hazards

Silicone based materials have a viscosity factor that can substantially reduce or eliminate traction. Be aware that improper applications of such products can cause serious injury or death. Please use warning labels on consumer products.

Signal Word:

WARNING

Physical Hazards:

No known physical hazards.

Acute Health Effects:

Route of Entry or Possible Contact:

Eyes, skin, inhalation (aerosol), ingestion

Eye Contact:

Causes severe eye irritation.

Skin contact:

Causes skin irritation. May be absorbed through skin.

Inhalation:

Aerosols prepared from aqueous emulsions or organic solutions of aminoalkyl-functional silanes may be dangerous to health referring to animal tests. See Sect. 3.3 "Chronic health effects".

Ingestion:

No known ingestion hazards.

Additional information on acute health effects:

The health hazard evaluation is based on tests results and/or on known properties of ingredients. Ingestion is not expected during industrial use.

3.3 Further Information:

Chronic health effects:

Prolonged or repeated inhalation of vapors may have adverse effects on the reproductive system, based on animal testing of a component of this material.

Medical conditions which may be aggravated by exposure:

None known.

Target organs affected:

Liver and Female Reproductive System.

Sign and Symptoms of Exposure:

Refer to Acute Health Effects, listed above.

Carcinogens/Reproductive Toxins:

Based on animal tests. This material contains between 0.1% and 1% of a known reproductive toxin. This material contains $\geq 0.1\%$ of a substance which may cause cancer. However, the relevance to humans has not been determined.

Section IV. First Aid Measures

- 4.1 General Information:**
Get medical attention if irritation occurs or if breath becomes difficult.
Remove contaminated clothing and shoes.
- 4.2 After Inhalation:**
If inhaled remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult give oxygen.
- 4.3 After contact with the skin:**
For skin contact, immediately wipe away excess material. Use a waterless hand cleaner to remove as much of the remaining material as possible. Wash with soap and water.
- 4.4 After contact with eyes:**
If contact with eyes, immediately hold eyelids apart and flush with plenty of water for at least 15 min.
- 4.5 After swallowing:**
For ingestion, if conscious, give several glasses of water but do not induce vomiting.
If vomiting does occur, give additional fluids.
- 4.6 Advice for the physician:**
Treat symptomatically.

Section V. Fire-fighting Measures

- | 5.1 Flammable Properties: | Method |
|-------------------------------|------------------|
| Flash Point | : not applicable |
| Boiling point / boiling range | : not determined |
| Lower explosion limit (LEL) | : not applicable |
| Upper explosion limit (UEL) | : not applicable |
| Ignition temperature | : not applicable |
- 5.2 Fire and explosion hazards:**
Material does not burn. This material does not present any unusual fire or explosion hazards.
- 5.3 Recommended extinguishing media:**
Carbon dioxide. Water – Use Fine Spray or Fog. Dry chemical.
- 5.4 Unsuitable extinguishing media:**
None.
- 5.5 Special exposure hazards arising from the substance or preparation itself, combustion products, resulting gases:**
Hazardous decomposition products: Various hydrocarbons fragments, carbon dioxide, formaldehyde, carbon monoxide, silicon dioxide.
- 5.6 Fire fighting procedures:**
Full turn-out gear and Self Contained Breathing Apparatus (SCBA) should be worn when fighting large fires.

Section VI. Accidental Release Measures

- 6.1 Precautions:**
Wear personal protection equipment (see section 8). Avoid inhaling mists and vapors.
If material is released indicate risk of slipping.
HAZWOPER PPE Level: C
- 6.2 Containment:**
Prevent material from entering surface waters, drains or sewers and soil. Contain any fluid that runs out using suitable material (e.g. earth). Retain contaminated water/extinguishing water.
- 6.3 Methods for Cleaning up:**
For small amounts: Absorb with a liquid binding material such as diatomaceous earth and dispose of according to local/state/federal regulations. Contain larger amounts and pump up into suitable containers.



Section VII. Handling and Storage

7.1 General Information:

Always stir well before use.

7.2 Handling

Precautions for safe handling:

Avoid formation of aerosols. In case of aerosol formation special protective measures are required (exhausting by suction, respiratory protection). Ensure adequate ventilation. Spilled substance increases risk of slipping.

Precautions against fire and explosion:

No special precautions against fire and explosion required.

7.3 Storage

Conditions for storage rooms and vessels:

Protect against frost.

Advice for storage of incompatible materials: not applicable

Further information for storage:

Protect against sun. Keep container tightly closed and store in a cool, well ventilated place.

Minimum temperature allowed during storage and transportation: 0 °C (32 °F)

Do not allow this material to freeze.

Maximum temperature allowed during storage and transportation: 32 °C (89 °F)

Temperature limit to maintain product quality.

Section VIII. Exposure Controls and Personal Protection

8.1 Engineering Controls

Ventilation:

No special ventilation required.

Local exhaust:

Local exhaust ventilation which meets the requirements of ANSI Z9.2 is recommended to control air-borne contaminants at the point of use.

8.2 Associate substances with specific control parameters such as limit values

Further Information:

Maximum concentration at workplace recommended by producer: octamethylcyclotetrasiloxane (D4, CAS no. 556-67-2) = 10 ppm (123 mg/m³)

8.3 Personal Protection Equipment (PPE)

Respiratory Protection:

A NIOSH approved air purifying respirator equipped with universal multi-contaminant, multi-gas/vapor cartridges and at least P-99 solid/aerosol particulate filters is recommended if overexposure to dusts, mists, or vapors could occur.

Hand protection:

Neoprene or nitrile rubber gloves

Eye protection:

Safety glasses with side shields. Additional eye and face protection, splash-proof goggles, hood, full-faced respirator, or face shield is recommended if splashing could occur.

Other protective clothing or equipment:

Provide eye bath and safety shower. Additional skin protection, such as SARANEX coated Tyvek apron, over-sleeves, lab coat, coveralls, or protective suit should be worn if splashing could occur.

8.4 General Hygiene and Protection Measures:

Wash thoroughly after handling. Avoid contact with eyes, skin and clothing.

Section IX. Physical and Chemical Properties

9.1 Appearance

Physical State/Form	: liquid
Color	: white
Odor	: weak

9.2 Safety Parameters

Melting point / melting range	: not determined
Boiling point / boiling range	: not determined
Flash point	: not applicable
Ignition temperature	: not applicable
Lower explosion limit (LEL)	: not applicable
Upper explosion limit (UEL)	: not applicable
Vapor Pressure	: not determined
Density	: 1.01 g/cm ³
Water Solubility / Miscibility	: Slightly Soluble
pH-Value	: not determined
Viscosity (dynamic)	: not determined

9.3 Further Information

Percent Volatiles	: 37%
VOC	: < 1.0 g/l

Maximum concentration at workplace recommended by producer: octamethylcyclotetrasiloxane (D4, CAS no. 556-67-2) = 10 ppm (123 mg/m³)

Section X. Stability and Reactivity

10.1 General Information:

If stored and handled in accordance with standard industrial practices no hazardous reaction are known.

10.2 Conditions to avoid:

None known.

10.3 Materials to avoid:

None known.

10.4 Hazardous decomposition products:

If stored and handled in accordance with standard industrial practices and local regulations where applicable:

None known.

The following applies for the silicone content of the substance:

Measurements have shown the formation of small amounts of formaldehyde at temperatures above about 150 °C (302 °F) through oxidation.

10.5 Further information:

Hazardous polymerization cannot occur.

Section XI. Toxicological Information

11.1 General information:

The toxicology information listed below is based on the components of the material.

11.2 Toxicological data:

Acute toxicity (LD50/LC50-values relevant to classification):

Exposition	Value/value range	Species	Source
by inhalation	> 1.4 mg/l/4h (spray / dust)	rat (Limit Test)	test report

Additional information / remarks:

Inhalation toxicity: In a female rat gender-specific range finding study (inhalation exposure) with OMCTS/D4 decreases in mean live litter size and in the number of implantation sites were seen at the 700 ppm exposure level.

Toxicity to reproduction/fertility: In a two generation reproduction study via inhalation with OMCTS/D4 in rats, decreased mean live litter size and prolonged labor (dystocia) were observed at the 500 ppm and 700 ppm exposure levels. These same effects were not seen at the lower dose levels of 70 ppm and 300 ppm. Previous inhalation developmental studies did not show birth defects at doses ranging up to 700 ppm in rats and 500 ppm in rabbits. The significance of these effects in humans cannot be determined at this time. Birth defects were only observed at maternally toxic doses. Because these effects are only seen at very high exposure levels, it is unlikely that industrial, commercial and/or consumer uses of products containing OMCTS/D4 would result in a significant risk to humans.

Chronic toxicity / carcinogenicity: In a two year combined chronic toxicity and carcinogenicity inhalation study with octamethylcyclotetrasiloxane (OMCTS/D4) in rats, an increased incidence of (uterine) endometrial cell hyperplasia and endometrial adenomas were observed at the highest exposure level of 700 ppm in female rats. These same effects were not seen at the other dose levels of 10, 30, and 150 ppm. Since these effects only occurred at 700 ppm, a level that greatly exceeds typical workplace or consumer exposure, it is unlikely that industrial, commercial or consumer uses of products containing OMCTS/D4 would result in a significant risk to humans.

Section XII. Ecological Information

12.1 Information on elimination (persistence and degradability)

Biodegradation / further information:

Silicone content: Biologically not degradable. Emulsifier: Product potentially degradable.

Further Information:

Silicone content: Elimination by adsorption to activated sludge

12.2 Behavior in environmental compartments

Mobility:

No data known.

Further Information:

No data known.

12.3 Ecotoxicological effects:

According to past experience toxicity to fish is improbable.

Effects in sewage treatment plants (bacteria toxicity: respiration/reproduction inhibition):

According to current knowledge adverse effects on water purification plants are not expected.

12.4 Additional Information:

Other harmful effects:

None known.

General Information:

According to our present knowledge no further data known.

Section XIII. Disposal Considerations

13.1 Product Disposal

Recommendation:

Dispose of according to regulations by incineration in a special waste incinerator. Observe local/state/federal regulations.

13.2 Packaging Disposal

Recommendation:

Completely discharge containers (no tear drops, no powder rest, scraped carefully). Containers may be recycled or re-used. Observe local/state/federal regulations.

Section XIV. Transport Information

14.1 US DOT & CANADA TDG SURFACE

Valuation : Not regulated for transport

Other Information : Protect from freezing. Temperature Sensitive Material.

Corrosive to Steel of Aluminum : Not corrosive to steel or aluminum.

14.2 Transport by sea IMDG-Code

Valuation : Not regulated for transport

14.3 Air Transport ICAO-TI/IATA-DGR

Valuation : Not regulated for transport

Section XV. Regulatory Information

15.1 U.S. Federal Regulations

TSCA Inventory Status and TSCA Information:

This material or its components are listed on or are in compliance with the requirements of the TSCA Chemical Substance Inventory.

TSCA 12(b) Export Notification:

This material does not contain any TSCA 12(b) regulated chemicals.

CERCLA Regulated Chemicals:

This material does not contain any CERCLA regulated chemicals.

SARA 302 EHS Chemicals:

This material does not contain any SARA extremely hazardous substances.

SARA 311/312 Hazard Class:

Immediate (acute) health hazard.

SARA 313 Chemicals:

This material does not contain any SARA 313 chemicals above de minimus levels.

HAPS (Hazardous Air Pollutants):

123-91-1 1,4-Dioxane

75-21-8 Ethylene oxide

15.2 U.S. State Regulations

California Proposition 65 Carcinogens:

123-91-1 1,4-Dioxane

75-21-8 Ethylene oxide

California Proposition 65 Reproductive Toxins:

75-21-8 Ethylene oxide

Massachusetts Substance List:

This material contains no listed components.

New Jersey Right-to-Know Hazardous Substance List:

This material contains no listed components.

Pennsylvania Right-to-Know Hazardous Substance List:

This material contains no listed components.

15.3 Canadian Regulations

This product has been classified in accordance with the Hazard criteria of the CPR and the MSDS contains all the information required by the CPR.

WHMIS Hazard Classes:

D2A, D2B

DSL Status:

This material or its components are listed on the Canadian Domestic Substances List.

Non-DSL Chemicals:

This material does not contain any non-DSL chemicals.

Canadian Ingredient Disclosure List:

9036-19-5 Polyoxyethylene (t-octylphenyl) ether

15.4 Other International Regulations

EU Risk Phrases:

R-Phrase	Description
R-	-

EU Safety Phrases:

S-Phrase	Description
S-	-

Details of international registration status listed on or in accordance with the following inventories:

IECSC - China
 TSCA - USA
 PICCS - Philippines
 ENCS - Japan
 ECL - Korea
 DSL - Canada
 AICS - Australia
 EINECS - Europe

Section XVI. Other Information

16.1 Additional Information:

This Material Safety Data Sheet (MSDS) meets the requirements of the Federal OSHA Hazard Communication Standard (29 CFR 1910.1200). This product has been classified according to the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all of the information required by the CPR. This information relates to the specific material designated and may not be valid for such material used in combination with any other materials or in any process. Such information is to the best of our knowledge and belief accurate and reliable as of the date compiled. However, no representation, warranty or guarantee expressed or implied, is made as to its accuracy, reliability or completeness. It is the user's responsibility to satisfy himself as to the suitability and completeness of such information for his own particular use. We do not accept liability for any loss or damage that may occur from the use of this information. Nothing herein shall be construed as a recommendation for uses which infringe valid patents or as extending a license under valid patents. This MSDS provides selected regulatory information on this product, including its components. This is not intended to include all regulations. It is the responsibility of the user to know and comply with all applicable rules, regulations and laws relating to the product being used.

16.2 Glossary of Terms:

ACGIH - American Conference of Governmental Industrial Hygienists
DOT - Department of Transportation
hPa - Hectopascals
mPa*s - Milli Pascal-Seconds
OSHA - Occupational Safety and Health Administration
PEL - Permissible Exposure Limit

ppm - Parts per Million
SARA - Superfund Amendments and Reauthorization Act
STEL - Short Term Exposure Limit
TSCA - Toxic Substances Control Act
TWA - Time Weighted Average
WHMIS - Canadian Workplace Hazardous Materials Identification System

Flash Point Determination Methods

ASTM D56
ASTM D92, DIN 51376, ISO 2592
ASTM D93, DIN 51758, ISO 2719
ASTM D3278, DIN 55680, ISO 3679
DIN 51755

Common name

Tagliabue (Tag) closed cup
Cleveland open cup
Pensky-Martens closed cup
Setaflash or Rapid closed cup
Abel-Pensky closed cup

16.3 Conversion table:

Pressure: 1 hPa * 0.75 = 1 mm Hg = 1 Torr; 1 bar = 1000 hPa
Viscosity: 1 mPa*s = 1 Centipoise (Cp)